

Cognition variables information.

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Summary

Cognitive function childhood, adolescence and early adulthood

Cognitive function was first tested at age 8 years in a school setting, and again in this setting at ages 11 and 15 years. A test of verbal comprehension given at the latter age was then re-administered at age 26 years (with 10 additional items of increased difficulty to avoid a ceiling effect), as part of the first interview of the survey member as an adult.

COG8H, COG11H and COG15H are summary measures of test scores standardised to the whole population on each component test. If using a sub-sample of the data these scores must be re-standardised to the population being analysed.

Cognitive function in adulthood

Apart from the reading comprehension test which was repeated at age 26 years, cognitive function was not assessed again until age 43 years. At this time there was a shift away from tests of intellectual ability and towards the assessment of functional cognitive performance (memory, processing speed and motor praxis).

Some of these tests were repeated at age 53 years (1999), with the addition of tests measuring verbal ability, verbal fluency, prospective memory and delayed verbal memory. The NART is the National Adult Reading Test and allows a link back to related tests in childhood.

Raw variables

VOC8R, VOC11R

Purpose : Score from a 50-item vocabulary test **at 8, 11 years** Year : 1954, 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

VOC8N, VOC11N

Purpose : Normalised score from a 50-item vocabulary test **at 8, 11 years** Year : 1954, 1957

Definition

Variables voc8r, voc11r normalised to a mean of 100 and SD 15

999 Missing data

SC8R

Purpose : Score from a 35-item verbal comprehension (sentence completion) test in **1954 (at 8 years)** Year : 1954

Definition

This variable is coded as a continuous variable with score values coded from 0 to 30

99 Missing data

SC8N

Purpose : Normalised score from a 35-item verbal comprehension (sentence completion) test in **1954 (at 8 years)** Year : 1954

Definition

Variable sc8r normalised to a mean of 100 and SD of 15

999 Missing data

R8R, R11R

Purpose : Score from a mechanical reading (pronunciation) test **at 8, 11 years** Year : 1954, 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

R8N, R11N

Purpose : Normalised score from a mechanical reading (pronunciation) test **at 8, 11 years** Year : 1954, 1957

Definition

Variable R8/11R normalised to a mean of 100 and SD of 15

999 Missing data

PI8R

Purpose : Score from a 60-item non-verbal ability (picture intelligence) test in **1954 (at 8 years)** Year : 1954

Definition

This variable is coded as a continuous variable with score values coded from 0 to 60

99 Missing data

PI8N

Purpose : Normalised score from a 60-item non-verbal ability (picture intelligence) test in **1954 (at 8 years)** Year : 1954

Definition

Variable PI8R normalised to a mean of 100 and SD of 15

999 Missing data

COG8H

Purpose : Standardised summary cognitive score in **1954 (at 8 years)** Year : 1954

Definition

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

2. Specification

COMPUTE COG8H=(VOC8H+SC8H+R8H+PI8H)/4 Where VOC8H, SC8H, R8H, PI8H are the test scores standardised to the whole population available on each test in the variables VOC8R, SC8R, R8R, PI8R.

COMPUTE VOC8H=(VOC8R-16.28)/5.99 *Xh=(Xr-mean)/SD

GA11R

Purpose : Score from a 80-item verbal and non-verbal ability test in **1957 (at 11 years)** Year : 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 80

99 Missing data

GA11N

Purpose : Normalised score of the 80-item verbal and non-verbal ability test in 1957 (at 11 years) Year : 1957

Definition

Variable GA11R normalised to a mean of 100 and SD of 15

999 Missing data

V11R

Purpose : Score from a 40 verbal items test in 1957 (at 11 years) Year : 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 40

99 Missing data

V11N

Purpose : Normalised score of the 40 verbal items in test v11r in 1957 (at 11 years) Year : 1957

Definition

Variable V11R normalised to a mean of 100 and SD of 15

999 Missing data

NV11R

Purpose : Score from a 40 non-verbal items test in 1957 (at 11 years) Year : 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 40

99 Missing data

NV11N

Purpose : Normalised score of the 40 non-verbal items in test v11r in 1957 (at 11 years) Year : 1957

Definition

Variable NV11R normalised to a mean of 100 and SD of 15

999 Missing data

A11R

Purpose : Score from a 50-item arithmetic test in 1957 (at 11 years) Year : 1957

Definition

This variable is coded as a continuous variable with score values coded from 0 to 50

99 Missing data

A11N

Purpose : Normalised score of the 50-item arithmetic test a11r in 1957 (at 11 years) Year : 1957

Definition

Variable A11R normalised to a mean of 100 and SD of 15

999 Missing data

COG11H

Purpose : Standardised summary cognitive score in 1957 (at 11 years) Year : 1957

Definition

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

2. Specification

COMPUTE COG11H=(R11H+VOC11H+GA11H+A11H)/4 Where R11H, VOC11H, GA11H, A11H are the test scores standardised to the whole population available on each test from the variables R11R, VOC11R, GA11R, A11R. COMPUTE R11H=(R11R-36.43)/10.51 *Xh=(Xr-mean)/SD

GA15R

Purpose : Score from the Heim AH4 Group Ability Test [a 130-item verbal and non-verbal ability test] in 1961 (at 15 years) Year : 1961

Definition

This variable is coded as a continuous variable with score values coded from 0 to 130

99 Missing data

GA15N

Purpose : Normalised score of the Heim AH4 Group Ability Test [a 130-item verbal and non-verbal ability test] in 1961 (at 15 years) Year : 1961

Definition

Variable GA15R normalised to a mean of 100 and SD of 15

999 Missing data

V15R

Purpose : Score from the 65 verbal items of the Heim AH4 Group Ability Test in 1961 (at 15 years) Year : 1961

Definition

This variable is coded as a continuous variable with score values coded from 0 to 65

99 Missing data

V15N

Purpose : Normalised score of the 65 verbal items of the Heim AH4 Group Ability Test in 1961 (at 15 years) Year : 1961

Definition

Variable V15R normalised to a mean of 100 and SD of 15

999 Missing data

NV15R

Purpose : Score from the 65 non-verbal items of the Heim AH4 Group Ability Test in 1961 (at 15 years) Year : 1961

Definition

This variable is coded as a continuous variable with score values coded from 0 to 65

99 Missing data

NV15N

Purpose : Normalised score of the 65 non-verbal items of the Heim AH4 Group Ability Test in 1961 (at 15 years)
Year : 1961

Definition

Variable NV15R normalised to a mean of 100 and SD of 15

999 Missing data

R15R

Purpose : Score from the Watts Vernon Reading Test [a 35-item verbal comprehension (sentence completion) test] in 1961 (at 15 years) Year : 1961

Definition

This variable is coded as a continuous variable with score values coded from 0 to 35

99 Missing data

R15N

Purpose : Normalised score of the Watts Vernon Reading Test [a 35-item verbal comprehension (sentence completion) test] in 1961 (at 15 years) Year : 1961

Definition

Variable GA15R normalised to a mean of 100 and SD of 15

999 Missing data

M15R

Purpose : Score from a 47-item mathematics test in 1961 (at 15 years) Year : 1961

Definition

This variable is coded as a continuous variable with score values coded from 0 to 47

99 Missing data

M15N

Purpose : Normalised score of the 47-item mathematics test in 1961 (at 15 years) Year : 1961

Definition

Variable NV15R normalised to a mean of 100 and SD of 15

999 Missing data

COG15H

Purpose : Standardised summary cognitive score in 1961 (at 15 years) Year : 1961

Definition

Summary variable of all standardised scores of variables VOC8R, SC8R, R8R, PI8R

2. Specification

COMPUTE COG15H=(R15H+ GA15H+M15H)/3 Where R15H, GA15H, M15H are the test scores standardised to the whole population available on each test from the variables R15R, GA15R, M15R. COMPUTE R11H=(R15R-24.47)/6.64 *Xh=(Xr-mean)/SD

WV26R

Purpose : Score on Watts Vernon Reading Test (first 35 items) used at 15 years in 1972 (**at 26 years**) Year : 1972

Definition

Watts Vernon Reading Test used at 15 years (in 1961). This variable is coded as a continuous variable with score values coded from 0 to 35

99 Missing data

R26R

Purpose : Score on Watts Vernon Reading Test used at 15 in 1972 (**at 26 years**) Year : 1972

Definition

Watts Vernon Reading Test used at 15 years (in 1961), with an additional 10 items of increased difficulty to avoid a ceiling effect. This variable is coded as a continuous variable with score values coded from 0 to 45

99 Missing data

R26N

Purpose : Score on Watts Vernon Reading Test used at 15 in 1972 (**at 26 years**) Year : 1972

Definition

Variable R26R normalised to a mean of 100 and SD of 15

999 Missing data

WV26H

Purpose : Standardised score on Watts Vernon Reading Test (first 35 items) used at 15 years in 1972 (**at 26 years**) Year : 1972

COMPUTE WV26H=(WV26R-28.76)/5.4 *Xh=(Xr-mean)/SD

R26H

Purpose : Standardised score on Watts Vernon Reading Test used at 15 in 1972 (**at 26 years**) Year : 1972

COMPUTE R26H=(R26R-32.79)/8.26 *Xh=(Xr-mean)/SD

ADULTHOOD (age 43 and 53) cognitive variables

mempr89, membp89, memlg89, memht89, memwt89, memac89, memcc89, memab89

Purpose : Memory for 8 medical measures* taken in 1982 when interviewed in 1989 (at 43 years) Year : 1989

Definition Remembered that pulse rate* was measured at 1982 interview *pulse rate (pr), blood pressure (bp), lung function (lg), height (ht), weight (wt), arm circumference (ac), chest circumference (cc) and abdominal circumference (ac). This variable is coded as 0 Did not remember [...] measured 1 Remembered [...] measured 7 No questionnaire 8 Not applicable 9 Unknown

Specification

DO IF (RANGE(BATCH89,1,18)) *if rcvd a 1989 questionnaire If (sysmis(memab89)) memab89=8 END IF

Note: these vars. have been subsequently labelled and an additional code '7' No questionnaire added. MR's original vars. were unlabelled and all those without a questionnaire in 1989 (N=2100) were blank (sysmis).

Variables in this document

25 medium-confidence matches

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory test [10135]		
memab89	Remembered that abdominal circumference was measured at 1982 interview - at age 43 years	prose scan
memac89	Remembered that arm circumference was measured at 1982 interview - at age 43 years	prose scan
membp89	Remembered that blood pressure was measured at 1982 interview - at age 43 years	prose scan
memcc89	Remembered that chest circumference was measured at 1982 interview - at age 43 years	prose scan
memht89	Remembered that height was measured at 1982 interview - at age 43 years	prose scan
memlg89	Remembered that lung function was measured at 1982 interview - at age 43 years	prose scan
mempr89	Remembered that pulse rate was measured at 1982 interview - at age 43 years	prose scan
memwt89	Remembered that weight was measured at 1982 interview - at age 43 years	prose scan
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Reading comprehension (Watts-Vernon) [10121]		
wv26r	Watts Vernon Reading Test (first 35 items) used at age 15 (edited raw score) - at age 26 years	prose scan
Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]		
cog11h	Cognitive function score (standardised to whole population on each test) - at age 11 years	prose scan
cog15h	Cognitive function score (standardised to whole population on each test) - at age 15 years	prose scan
cog8h	Cognitive function score (standardised to whole population on each test) - at age 8 years	prose scan
ga11n	Verbal and non-verbal ability - General Ability (80-items - normalised score) - at age 11 years	prose scan
ga11r	Verbal and non-verbal ability - General Ability (80-items - edited raw score) - at age 11 years	prose scan
ga15n	Heim A4 Group ability verbal and non-verbal test (130-items - normalised score) - at age 15 years	prose scan
ga15r	Heim A4 Group ability verbal and non-verbal test (130-items - edited raw score) - at age 15 years	prose scan
nv11n	Non-verbal intelligence test (40-items normalised score) - at age 11 years	prose scan
nv11r	Non-verbal intelligence test (40-items edited raw score) - at age 11 years	prose scan
nv15n	Heim A4 Group ability non-verbal test (65-items - normalised score) - at age 15 years	prose scan
nv15r	Heim A4 Group ability non-verbal test (65-items - edited raw score) - at age 15 years	prose scan
voc11n	Vocabulary test (50-items normalised score) - at age 11 years	prose scan

Variable	Label	How it was found
voc11r	Vocabulary test (50-items edited raw score) - at age 11 years	prose scan
voc8n	Vocabulary test (50-items normalised score) - at age 8 years	prose scan
voc8r	Vocabulary test (50-items edited raw score) - at age 8 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan

22 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Reading comprehension (Watts-Vernon) [10121]		
r26n	Watts Vernon Reading Test used at age 15 (normalised score) - at age 26 years	prose scan
r26r	Watts Vernon Reading Test used at age 15 (edited raw score) - at age 26 years	prose scan
Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]		
a11n	Arithmetic test (50-items - normalised score) - at age 11 years	prose scan
a11r	Arithmetic test (50-items - edited raw score) - at age 11 years	prose scan
m15n	Mathematics test (47-items - normalised score) - at age 15 years	prose scan
m15r	Mathematics test (47-items - edited raw score) - at age 15 years	prose scan
pi8n	Non-verbal ability - picture intelligence (60-items - normalised score) - at age 8 years	prose scan
pi8r	Non-verbal ability - picture intelligence (60-items - edited raw score) - at age 8 years	prose scan
r11n	Mechanical reading - pronunciation - (normalised score) - at age 11 years	prose scan
r11r	Mechanical reading - pronunciation - (edited raw score) - at age 11 years	prose scan
r15n	Watts Vernon Reading Test (35-items verbal comprehension normalised score) - at age 15 years	prose scan
r15r	Watts Vernon Reading Test (35-items verbal comprehension edited raw score) - at age 15 years	prose scan
r8n	Mechanical reading - pronunciation - (normalised score) - at age 8 years	prose scan
r8r	Mechanical reading - pronunciation - (edited raw score) - at age 8 years	prose scan
sc8n	Verbal comprehension - sentence completion (35-items normalised score) - at age 8 years	prose scan
sc8r	Verbal comprehension - sentence completion (35-items edited raw score) - at age 8 years	prose scan
v11n	Verbal Intelligence (40-items - normalised score) - at age 11 years	prose scan
v11r	Verbal Intelligence (40-items - edited raw score) - at age 11 years	prose scan
v15n	Heim A4 Group ability verbal test (65-items - normalised score) - at age 15 years	prose scan
v15r	Heim A4 Group ability verbal test (65-items - edited raw score) - at age 15 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Physical capability [5108]		
arm	Do you have difficulty because of long term health problems using either arm to reach up high above your head or to reach behind to tuck a shirt in or do up a zip?- at age 53 years	prose scan